WHEN IS AN INDUCTION PRODUCT OF 0-HECKE MODULES DECOMPOSABLE? THE ONE-BOX CRITERION (Q74)

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This PDF corresponds to work-in-progress source commit `fb9badb`.
Follow-up to the Q72(i)/Q73 note (work-in-progress commit `3ab7f44`).
Proof record (local): `proofs/2026-11-11-q74-gstar-box-criterion.md` (PROVE 291, cold read in Dream 290),
research-repo commit `fffbdd9`; registry `proofs/registry/q74-gstar-failure.json`.

Summary. For indecomposable 0-Hecke modules X, Y with multiplicity-free tops, the induction product $X \boxtimes Y$ is decomposable if and only if $X \boxtimes \square$ and $\square \boxtimes Y$ both are, where $\square$ is the one-box module (Theorem 4). Under a hereditary version of the hypothesis, "$X \boxtimes \square$ decomposes" means exactly that $X \cong X' \boxtimes P$ with P a nonzero indecomposable projective (Proposition 6). This explains every failure, among pairs satisfying (H), of the conjecture (G*) ("$X \boxtimes Y$ decomposes only if both are projective"): the failures are associativity, $P \boxtimes (\square \boxtimes S) \cong (P \boxtimes \square) \boxtimes S$.
Grades. Theorem 4 and Proposition 6: proved (internal grade; one internal cold read, PASS, 0 must-fix). The table in §4: computed. That (H_∞) holds for all weak-Bruhat interval modules: hunch. Owner check (2026-11-12): not found; see §5.
What I want from you: a referee-style check of Step B (sufficiency), and a pointer if you know of a source.

1. Setting and hypotheses

$\Bbbk$ is a field and $H_n = H_n(0)$ is generated by $\pi_1, \dots, \pi_{n-1}$ with $\pi_i^2 = \pi_i$ and the braid relations. For $D \subseteq [n-1]$, C_D is the one-dimensional module with $\pi_i \mapsto [i \in D]$. For $X \in \operatorname{mod} H_a$ and $Y \in \operatorname{mod} H_b$, with $N = a + b$ and $B = H_a \otimes H_b \subseteq H_N$, put

$$X \boxtimes Y := \operatorname{Ind}_B^{H_N}(X \otimes Y), \qquad \square := C_\emptyset \in \operatorname{mod} H_1.$$

We use the word model of the Q72/Q73 note: $X \boxtimes Y$ has basis-free decomposition $\bigoplus_w T_w \otimes (X \otimes Y)$ over (a,b)-shuffles w, encoded by value words in $\{A, B\}^N$, and $\operatorname{Res}_B(X \boxtimes Y) = \bigoplus_k M_k$ (Mackey), where k is the number of B's among the first a letters and $M_0 \cong X \otimes Y$.

Definition 1. (H) for a pair (X, Y): X and Y are indecomposable with multiplicity-free tops $\operatorname{top} X = \bigoplus_i C_{D_i}$, $\operatorname{top} Y = \bigoplus_j C_{E_j}$. **($\mathbf{H}_\infty$)** for X: for every $0 \le t \le a$, every indecomposable summand of $\operatorname{Res}_{H_{a-t}} X$ (via $\pi_1, \dots, \pi_{a-t-1}$) has a multiplicity-free top.

We use three results from Part II of the Q72/Q73 note, all proved there for arbitrary modules (no shapes): Lemma II.3 (the top of $X \boxtimes Y$ is multiplicity-free, with labels $Z_{ij0} = D_i \cup (a + E_j)$ and $Z_{ij1} = Z_{ij0} \cup \{a\}$, functionals f_{ije}); Lemma II.4 (criterion: under (H), $X \boxtimes Y$ is decomposable iff there is a *witness* $k \ge 1$, (i, j), $u \in \operatorname{Hom}_B(M_0, M_k)$ with $f_{ij1}(u(x_i \otimes y_j)) \ne 0$); Lemma II.5 (Key Lemma). Here x_i, y_j are top vectors with $\varepsilon_i(x_i) = 1 = \delta_j(y_j)$.

Definition 2. For $1 \le k \le a$ let $w_0^{X,k}$ be the longest element of $\langle s_{a-k+1}, \dots, s_{a-1} \rangle$, and

$$\widehat{L}_k(X) := \operatorname{Ind}_{H_{a-k} \otimes H_k}^{H_a}\big(\operatorname{Res}_{H_{a-k}} X \boxtimes C_{[k-1]}\big), \qquad R_k(X) :\iff X \text{ is a direct summand of } \widehat{L}_k(X).$$

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Mirror: $\widehat{L}'_k(Y) := \mathrm{Ind}_{H_k\otimes H_{b-k}}^{H_b}(C_{[k-1]}\boxtimes \mathrm{Res}'\, Y)$, with Res' via $\pi_{k+1},\dots,\pi_{b-1}$, and $L_k(Y) :\Longleftrightarrow\ Y \mid \widehat{L}'_k(Y)$. Note $C_{[k-1]} = P_{(k)}$ is the one-dimensional projective.

Fact 3 (Frobenius)**.** $\mathrm{Hom}_{H_a}(\widehat{L}_k(X), X) = \{\zeta_\theta\}$, *where* $\zeta_\theta(T_\ell\otimes x\otimes 1) = T_\ell\, \pi_{w_0^{X,k}}\theta(x)$ *and* θ *runs over* $\mathrm{End}_{H_{a-k}}(\mathrm{Res}\, X)$. *Indeed the joint fixed space of* $\pi_{a-k+1},\dots,\pi_{a-1}$ *is* $\pi_{w_0^{X,k}}X$, *and* $\pi_{w_0^{X,k}}$ *commutes with* $\pi_1,\dots,\pi_{a-k-1}$. *Write* $\zeta := \zeta_{\mathrm{id}}$*; then* $\zeta_\theta = \zeta\circ \mathrm{Ind}(\theta\boxtimes \mathrm{id})$.

2. The theorem

Theorem 4 (one-box criterion)**.** *Assume (H) for* (X,Y). *The following are equivalent:*

(1) $X\boxtimes Y$ *is decomposable;*
(2) $R_k(X)$ *and* $L_k(Y)$ *hold for some* $1\le k\le \min(a,b)$*;*
(3) $R_1(X)$ *and* $L_1(Y)$ *hold;*
(4) $X\boxtimes\square$ *and* $\square\boxtimes Y$ *are both decomposable.*

Proof sketch. **Step A, (1)⇒(2).** The Key Lemma II.5(b) uses only (H). A witness (k,i,j,u) yields maps $X\to \widehat{L}_k(X)\to X$ whose composite g has $\varepsilon_i\circ g\neq 0$, so $g\notin J(\mathrm{End}\, X) = \ker\rho^X$ and g is a unit: $R_k(X)$. For $L_k(Y)$ at the *same* k, twist by $\varphi:\pi_i\mapsto\pi_{N-i}$. Then $\varphi^*(X\boxtimes Y)\cong \varphi^*Y\boxtimes\varphi^*X$; shuffles go to shuffles, value words are reversed with $A\leftrightarrow B$, so the level k goes to $a-(a-k)=k$, and top labels go $Z\mapsto N-Z$, so $f_{ij1}\mapsto f'_{ji1}$. A level-k witness for (X,Y) is one for (φ^*Y,φ^*X), and $R_k(\varphi^*Y)$ twists back to $L_k(Y)$.

Step B, (2)⇒(1) (new). *Units.* By $R_k(X)$ and Fact 3 there is $\alpha':X\to\widehat{L}_k(X)$ with $\zeta\alpha'$ a unit of the local ring $\mathrm{End}\, X$; it acts on the multiplicity-free top by nonzero scalars, so $\varepsilon_i(\zeta\alpha'(x_i)) = c_i\neq 0$ for every i. Likewise $\beta':Y\to\widehat{L}'_k(Y)$ with $\delta_j(\zeta'\beta'(y_j)) = c'_j\neq 0$, where $\zeta'(T_r\otimes 1\otimes y) = T_r\pi_{w_0^{Y,k}}y$. *Structure of* M_k. With $B'' = H_{a-k}\otimes H_k\otimes H_k\otimes H_{b-k}\subseteq B$,

$$M_k\cong \mathrm{Ind}_{B''}^{B}\, V^{\mathrm{tw}},$$

where $V^{\mathrm{tw}} = X\otimes Y$ with H_{a-k} acting on X by $\pi_1..\pi_{a-k-1}$, the first H_k on Y by $\pi_1..\pi_{k-1}$, the second H_k on X by $\pi_{a-k+1}..\pi_{a-1}$, and H_{b-k} on Y by $\pi_{k+1}..\pi_{b-1}$. (The left word ℓ holds A-values $1..a-k$ and B-values $1..k$; the right word r holds the rest; AA/BB steps act inside each block and AB/BA are coset moves.) *The map.* $\psi: x\otimes 1\otimes 1\otimes y\mapsto \pi_{w_0^{X,k}}x\otimes\pi_{w_0^{Y,k}}y$ is B''-linear from $(\mathrm{Res}\, X\boxtimes C_{[k-1]})\otimes(C_{[k-1]}\boxtimes\mathrm{Res}'\, Y)$ to V^{tw}, factor by factor. Put $u := \mathrm{Ind}(\psi)\circ(\alpha'\otimes\beta')\in \mathrm{Hom}_B(M_0,M_k)$. *The witness.* By the region formula $f_{ij1}(T_{(\ell|r)}\otimes x\otimes y) = [\ell\ \mathrm{ok}_i][r\ \mathrm{ok}_j]\,\varepsilon_i(x)\delta_j(y)$,

$$f_{ij1}(u(x_i\otimes y_j)) = \varepsilon_i(\zeta\alpha'(x_i))\cdot\delta_j(\zeta'\beta'(y_j)) = c_i\,c'_j\neq 0.$$

(For the first factor: $\varepsilon_i(T_\ell\pi_{w_0}x) = \chi_{D_i}(T_\ell)\chi_{D_i}(\pi_{w_0})\varepsilon_i(x)$; since $c_i\neq 0$ we have $[a-k+1,a-1]\subseteq D_i$, and then $\chi_{D_i}(T_\ell) = [\ell\ \mathrm{ok}_i]$ as in Lemma II.5.) So u is a witness and Lemma II.4 gives (1).

Step C, (2)⇒(3); no hypothesis needed. Let $k\ge 2$. The vector $T_{s_1\cdots s_{k-1}}\otimes 1\otimes 1$ of $\mathrm{Ind}_{(k-1,1)}(C_{[k-2]}\boxtimes\square)$ has value word BA^{k-1}, the longest shuffle, so every π_i fixes it; this gives $C_{[k-1]}\to\mathrm{Ind}(C_{[k-2]}\boxtimes\square)$ and, by transitivity of induction, $\iota:\widehat{L}_k(X)\to\mathrm{Ind}_{(a-1,1)}(W\boxtimes\square)$ with $W = \mathrm{Ind}_{(a-k,k-1)}(\mathrm{Res}_{H_{a-k}}X\boxtimes C_{[k-2]})$. Since $s_1\cdots s_{k-1}\cdot w_0(S_{k-1}) = w_0(S_k)$ is length-additive,

$$\zeta = \zeta^{(1)}\circ\mathrm{Ind}(\zeta''\boxtimes\mathrm{id})\circ\iota,$$

with ζ'' the Fact 3 map for $W\to\mathrm{Res}_{H_{a-1}}X$ and $\zeta^{(1)}$ the $k=1$ map. So $\zeta\alpha'$ factors through $\widehat{L}_1(X)$ and $R_1(X)$ holds. The mirror follows by φ^*. (3)⇒(2) is trivial.

Step D, (3)⇔(4). Apply (1)⇔(2) to $(X,\square)$: $\square$ is simple, $b=1$ forces $k=1$, and $L_1(\square)$ holds since $\widehat{L}'_1(\square)=\square$. So $X\boxtimes\square$ is decomposable iff $R_1(X)$; mirror for Y. □

3. Consequences

Corollary 5. *(a) $R_1(P)$ and $L_1(P)$ hold for every nonzero projective P, since $P \boxtimes \square$ is projective with a two-element top (Lemma II.3).*

(b) The violators of (G) are* exactly *the pairs (X, Y) satisfying (H) with $R_1(X)$, $L_1(Y)$, and X or Y non-projective. In particular a non-projective X with $R_1(X)$ violates (G*) against every projective Y.*

(c) If $Y \cong Q \boxtimes Y'$ with Q a nonzero projective, then $L_1(Y)$, because $\square \boxtimes Y \cong (\square \boxtimes Q) \boxtimes Y'$ and $\square \boxtimes Q$ is a sum of two projectives (Huang; KLO Lemma 5.1).

Proposition 6 (structure of the special modules)**.** *Let X be indecomposable with (H_∞). Then $R_1(X)$ holds iff $X \cong X' \boxtimes P_\gamma$, where P_γ is an indecomposable projective with $|\gamma| \geq 1$ and X' is an indecomposable summand of $\mathrm{Res}_{H_{a-|\gamma|}} X$ ($X' = \Bbbk$ if X is projective). The mirror statement holds for L_1.*

Proof sketch. ($\Leftarrow$) $X \boxtimes \square \cong X' \boxtimes (P_\gamma \boxtimes \square) = X' \boxtimes P_{\gamma \cdot 1} \oplus X' \boxtimes P_{\gamma \odot 1}$, both nonzero; apply Theorem 4 to $(X, \square)$. This needs only (H) for X. ($\Rightarrow$) Keep the invariant $X \mid Z^{(t)} \boxtimes P_\gamma$ with $|\gamma| = t$ and $Z^{(t)}$ an indecomposable summand of $\mathrm{Res}_{H_{a-t}} X$. For $t = 1$: $X \mid \mathrm{Res}\, X \boxtimes \square = \bigoplus_\nu Z_\nu \boxtimes \square$; use Krull–Schmidt. If $Z^{(t)} \boxtimes P_\gamma$ is indecomposable, then $X \cong Z^{(t)} \boxtimes P_\gamma$ and we stop. Otherwise $(Z^{(t)}, P_\gamma)$ satisfies (H) (by (H_∞), and P_γ has simple top), so Theorem 4 gives $R_1(Z^{(t)})$, i.e. $Z^{(t)} \mid Z^{(t+1)} \boxtimes \square$; then $X \mid Z^{(t+1)} \boxtimes (\square \boxtimes P_\gamma) = Z^{(t+1)} \boxtimes P_{1 \cdot \gamma} \oplus Z^{(t+1)} \boxtimes P_{1 \odot \gamma}$ and Krull–Schmidt restores the invariant. At $t = a$, $Z^{(a)} = \Bbbk$ and $X \cong P_\gamma$. □

Corollary 7 (skew shapes; the seam form of KLO Conj. 7.5)**.** *Skew shape modules satisfy (H_∞): $\mathrm{Res}_{H_{a-t}} X_{\lambda/\mu} = \bigoplus_\nu X_{\nu/\mu}^{\oplus f^{\lambda/\nu}}$, and each $X_{\nu/\mu}$ splits into indecomposables X_S with multiplicity-free tops (Q73 strong form). For shapes S, T with X_S, X_T indecomposable,*

$$X_{S \star T} = X_S \boxtimes X_T \text{ is decomposable} \iff \begin{array}{l}\text{the last component of } S \text{ and} \\ \text{the first component of } T \text{ are ribbons.}\end{array}$$

Indeed $R_1(X_S) \iff X_{S \star (1)}$ decomposable $\iff$ the last component of S is a ribbon (Q73 strong form), and similarly for L_1. This agrees with Q73 (= KLO Conj. 7.5) and uses it only at the one-box level $S \star (1)$; by Proposition 6, $R_1(X_S)$ also means $X_S \cong X' \boxtimes P$.

4. Computations

Computed. Indecomposable weak-Bruhat interval modules of $H_m(0)$, $m \leq 5$; decomposability by a locality test of End over $\mathbb{F}_{32003}$ (scripts `q74.py`, `box.py`, `rl.py` in `proofs/prove291/`). All of them have multiplicity-free tops, so (H) holds for every pair.

m	1	2	3	4	5
indecomposable intervals	1	2	8	62	822
projective	1	2	4	8	16
R_1 (= $X \boxtimes \square$ dec.); same count for L_1	1	2	4	12	74
non-projective with R_1	0	0	0	4	58

Theorem 4(1)⇔(4), all pairs with $a + b \leq 8$, $\binom{a+b}{a} \dim X \dim Y \leq 1500$; cumulative in a

$a \leq$	1	2	3	4	5
pairs	895	2685	9777	14299	23273
decomposable	93	279	619	843	1329
mismatches	0	0	0	0	**0**

- The four non-projective R_1-modules at $m = 4$ (the "special" modules of the (G*) tally) are $\square \boxtimes S$ or $S \boxtimes \square$ with S a non-projective simple of H_3, e.g. $[1243, 4132] \cong \square \boxtimes C_{\{2\}}$ (isomorphisms

checked by random invertible elements of Hom). An earlier claim that no special modules occur at sizes 5–6 was an artifact of enumerating only $m \le 4$.

- At $m = 5$ the 58 non-projective R_1-modules coincide with the modules of iso-form $X' \boxtimes P$ (all 822 checked), in line with Proposition 6 beyond the cases where (H_∞) is proved.
- Iso-form criterion versus decomposability: 6417/6417 agree for $m \le 5$, $a + b \le 7$, dimension cap 4000.
- Every nonempty set $\{k : R_k(X)\}$ or $\{k : L_k(Y)\}$ contains $k = 1$ ($m \le 4$ all; $m = 5$ with dim ≤ 12), as Step C predicts.

5. Owner check

Not found (2026-11-12). The nearest prior result is Kim–Lee–Oh, arXiv:2310.20571 v2 (20 Jan 2025), Prop. 7.3(2): if $\mathrm{sh}(\tau_P)$ contains a disconnected ribbon then M_P is not indecomposable, proved via $P_\alpha \boxtimes P_\beta = P_{\alpha\cdot\beta} \oplus P_{\alpha\odot\beta}$ (Huang, Thm. 3.3). That is the special case of the sufficiency direction of Theorem 4 (and of Corollary 5(c)) in which both factors are ribbon, i.e. projective, modules. Conj. 7.5 is still stated as open in v2. Checked first-hand, with no one-box criterion and no analogue of Proposition 6: Jung–Kim–Lee–Oh 2105.14169; Choi–Kim–Nam–Oh 2003.11225, 2008.06830, 2201.13252; Choi–Kim–Oh 2308.10456; and all six citers of 2310.20571 (2609.10911, 2412.08413, 2410.07990, 2405.05867, 2311.10068, 2211.06575).

6. What I am least sure of

- **(H_∞) beyond skew shapes** is a hunch. For interval modules I checked multiplicity-free tops ($m \le 5$) but not the tops of the summands of all restrictions. Without (H_∞), the converse of Corollary 5(c) ("$L_1(Y) \Rightarrow Y \cong Q \boxtimes Y'$") is computed only.
- **Hypothesis (H) is essential to the method.** It enters only through Lemma II.4, whose map ρ needs multiplicity-free tops; I have no statement without it.
- **The proof rests on the Q72/Q73 note**, Lemmas II.3–II.5, which have had one internal cold read but no external check. The proof record marked three spots for rechecking (the identification of M_k with the twisted outer induction in Step B, the twist bookkeeping in Step A, transitivity of induction in Step C); the cold read checked all three, and an outside check of Step B is still the most useful.
- **The decomposability tests are modular** ($\mathbb{F}_{32003}$, randomised End computation) and subject to a dimension cap; they support but do not certify the characteristic-free statement.